

Technology Companion

Doing Statistics in Stats Without Walls

A hands-on companion to
You're Smarter Than You Think: Statistics

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Keyed chapter-by-chapter to the main book

*"The computer does the arithmetic. Your job is to ask the right question
and read the answer. You can absolutely do that."*

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Before You Click a Single Thing

If “statistics software” sounds like something that will crash, cost money, or expect you to know code, breathe. Stats Without Walls is none of those. It is free, forever. There is no coding and no account. You open a web page, you point, you click, and the results appear with the formula, a plain-English conclusion and a condition check under every one.

This companion follows the main book chapter for chapter. When you finish Chapter 4 in the book, open Chapter 4 here and watch the same ideas happen in Stats Without Walls. You never have to guess which menu goes with which idea.

One promise: every click-path in this book is short, and every one is followed by what the result looks like and what it means. You are never left staring at a panel wondering what you are seeing.

Getting Stats Without Walls

Stats Without Walls lives at learnwithoutwalls.com/stats. You have three easy choices, and they all give the same app:

1. **Open it in your browser.** Chrome, Edge, Safari or Firefox, on a laptop, a school computer or a tablet. Nothing to install. This is the version most people use, and it always has the newest features.
2. **Install it from the browser.** In Chrome or Edge, click the install icon in the address bar. It opens in its own window and works offline after the first visit.
3. **Download the desktop app** (Mac or Windows) for a computer that is often offline. The link is on the app's download page: learnwithoutwalls.com/stats/download.html.

Your data never leaves your computer: there is no server behind the page. When the app opens you will see two things. Those two things are your whole world:

- **The data table** on the left: the spreadsheet. This is where your numbers live. Under every column name the app prints the type it guessed: **nominal** for words, **continuous** for numbers.
- **The menu bar** across the top (Data, Exploration, Graph, T-Tests, ANOVA, Regression, Frequencies, distrACTION, and more). This is where you ask your questions. Results appear on the right, in the **Analyses** tab; graphs from the Graph menu go to the **Graphs** tab.

That is it. A place for the data, a place for the questions. Everything in this book happens in one of those two places. A full user guide is one click away in the app header.

Getting Data Into Stats Without Walls

Every example in this book comes as a .csv file (a plain spreadsheet). Opening one takes two clicks.

Open a file

Stats Without Walls (click this path)

Data > Open CSV file > pick the .csv file

The data fills the table on the left. That is the file loaded. (Data > Paste data works too, straight from a spreadsheet, and Data > Sample datasets has ready-made files.)

Then check your variable types. The app guesses a type for every column and prints it under the column name. The guess is usually right, but it is the one habit that makes everything else work, so look:

- **Nominal** or **Ordinal** for categories (words), like Home / Library / Cafe.
- **Continuous** for measurements (numbers), like a quiz score.

If a guess is wrong (a numeric code that is really a category, or a number column that was read as words because one cell holds text), fix it with Data > Set variable type. That is the “words vs. numbers” choice from Chapter 1, made official. The app uses it to decide which graphs and tests it will offer you. Now let’s walk the book.

Chapter 1

What's the Story in the Data?

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Get data into Stats Without Walls, check which variables it read as *words* (categories) and which as *numbers*, and look at your table without panic.

In Chapter 1 you learned the difference between a population and a sample, and between categorical and quantitative variables. In Stats Without Walls, that difference is a real setting printed under every column name, and it decides what the app will let you do later.

Stats Without Walls (click this path)

Data > Open CSV file > class.csv

Read the type under each column name:

"study_spot" nominal (leave it)

"quiz_score" continuous (leave it)

To change one: Data > Set variable type

What you'll see

study_spot	Nominal	(a category: Home / Library / Cafe)
quiz_score	Continuous	(a number the app can average)

■ READ THE OUTPUT

study_spot as **Nominal** tells the app these are categories, not text to average. quiz_score as **Continuous** tells the app these are numbers it can add up and average. That is exactly the “words vs. numbers” choice from the chapter, made official, and it is why later chapters just work.

■ YOUR TURN

Use Data > Compute or edit cells to add a column called `major` with five majors of your choice. What type does the app print under it? Now try Data > Set variable type on `quiz_score` and change it to Nominal: which menus stop offering it, and why?

Chapter 2

Where Did This Data Come From?

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Draw random samples from a population, see how they differ from each other, and learn what to check about a sample once you have one.

Chapter 2 is about how good data is collected. The gold standard is the simple random sample: everyone has an equal chance. Stats Without Walls lets you watch that happen. Load the forty students and let the app draw samples from them; every sample is different, and that difference is the whole reason the second half of the course exists.

Stats Without Walls (click this path)

```
Data > Open CSV file > population40.csv
```

```
Learn > Sampling distribution simulator
```

```
Population: a column of the data Column: gpa Statistic: mean
```

```
Sample size 10, Number of samples 1 (then try 1000)
```

What you'll see

```
Population N 40 mean (the truth you are trying to estimate)
```

```
Sample n 10 mean (this sample's guess; run it again and it changes)
```

■ READ THE OUTPUT

No software can promise your sample was random; only your *method* can do that (a random-number list, a lottery, a survey tool). What the app can do is draw honest random samples from a population you already have, so you can see sampling variability with your own eyes: run the simulator twice and the two sample means disagree. Then use Exploration > Descriptives on the full file to sanity-check the population: its size, its center, nothing wildly out of range.

■ YOUR TURN

Load `population40.csv`, run the simulator with sample size 10 and number of samples 1, three times, and write down the three sample means. Then run Descriptives on `gpa` for all 40 and write one sentence: how far were your three guesses from the truth, and why?

Chapter 3

Show Me a Picture

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Make the three core graphs from the chapter: a bar plot for categories and a histogram for numbers (plus a box plot).

Stats Without Walls uses your “words vs. numbers” choice to offer the right picture. Bar plots for categories, histograms for numbers. Everything lives under one analysis.

Stats Without Walls (click this path)

Exploration > Descriptives

select "study_spot" and "quiz_score" (hold Cmd or Ctrl for both)

tick under Plots:

Bar plot (for study_spot)

Histogram (for quiz_score)

Box plot (for quiz_score)

(or: Graph > Bar Plot, Graph > Histogram, Graph > Boxplot)

What you'll see

Bar plot: Home is the tallest bar (most common study spot)

Histogram: the bars pile up around 14 to 15

Box plot: a box over the middle 50% of the scores

■ READ THE OUTPUT

The **Bar plot** draws one bar per category with gaps, and Home is tallest, just like the book. The **Histogram** bins the numbers into touching bars, so you see the scores pile up around 14 to 15. The **Box plot** shows the middle, the spread, and any outliers. Each plot lands in the card with the count printed above every bar and bin, and the count axis always starts at zero. The Graph menu draws the same pictures in their own Graphs tab, with a frequency table underneath.

■ YOUR TURN

In the Plots section, look for the histogram's bin setting and change it. Do the bars get chunkier or finer? Does the "pile" still sit near 14 to 15?

Chapter 4

The Typical and the Spread

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Compute the center (mean, median) and the spread (standard deviation, IQR) with a few checkboxes.

Chapter 4's formulas are checkboxes in Stats Without Walls. You do not compute them by hand here; you check your by-hand work against the app, and the app prints the formula it used under the table so the two can be compared line by line.

Stats Without Walls (click this path)

```
Exploration > Descriptives (select "quiz_score")
open "Statistics" and check:
  Mean    Median    Std. deviation    IQR
```

What you'll see

Mean	14.0
Median	14.0
Std. deviation	1.15
IQR	1.25

■ READ THE OUTPUT

Stats Without Walls lays the numbers in a tidy column. Here the mean and median are both 14, which says the scores are fairly symmetric. If the mean were pulled well above the median, that would warn you of a right skew or a high outlier, exactly the reasoning from the chapter. The standard deviation, about 1.15, is the typical distance of a score from that mean.

■ YOUR TURN

Add one unusually high score, say 40, to the bottom of the `quiz_score` column and watch the Mean and Median update live. Which one moved more? That is why we report the median for skewed data.

Chapter 5

What Are the Chances?

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Build a two-way table, add its totals, and read probabilities and conditional probabilities straight off it.

Chapter 5's two-way tables are where probability gets concrete. Stats Without Walls turns counts into row percentages for you.

Stats Without Walls (click this path)

Frequencies > Contingency Tables > Independent Samples

Rows: caffeine

Columns: slept_well

open "Cells" and check: Row (for row percentages)

What you'll see

	Slept: Yes	Slept: No	Total
Caffeine Yes	30	20	50
Caffeine No	10	40	50
Total	40	60	100

■ READ THE OUTPUT

The counts and the **Total** row and column give you the whole picture: 40 of the 100 people slept well overall. Turning on **Row** percentages answers the conditional question: *given* they had caffeine, what fraction slept well? That is conditional probability, the trickiest idea in the chapter, made into a single checkbox.

■ YOUR TURN

In the Cells section, uncheck Row and check **Column** percentages instead. Now you are conditioning on whether they slept well. In plain words, what question does that table answer?

Chapter 6

Predictable Randomness

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Get exact binomial and normal probabilities without a table in the back of the book.

Chapter 6's distribution tables live in the **distrACTION** menu. Nothing to install: it is built in, and every answer comes with a picture of the distribution with the probability shaded.

Stats Without Walls (click this path)

```
distrACTION > Normal Distribution
```

```
Mean 70, SD 8:  Compute: below      x1 = 80      gives P(X < 80)
                  Compute: quantile  p = 0.90      gives the score at the 90th
                  percentile
```

```
distrACTION > Binomial Distribution
```

```
n 10, p 0.7:    P(X = 8)    and    P(X <= 8)
```

What you'll see

Normal	P(X < 80)	= 0.894
Normal	Quantile at 0.90	= 80.25
Binomial	P(X = 8)	= 0.233
Binomial	P(X <= 8)	= 0.851

■ READ THE OUTPUT

$P(X < 80) = 0.894$ says about 89% of scores fall below 80. The quantile runs it backward: to be in the top 10%, you need about 80.25. One box takes a score and gives a probability; the other takes a probability and gives a score. They are opposites. For the binomial, $P(X = 8) = 0.233$ is the chance of exactly 8 correct out of 10.

■ YOUR TURN

In Normal Distribution, find the probability a score is *above* 85. (Hint: choose Compute: above, or take 1 minus $P(X < 85)$. Check that the shaded part of the picture is the part you meant.)

Chapter 7

From Sample to Truth

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Watch the Central Limit Theorem happen, then read the standard error, which the app shows you directly.

Chapter 7 claims something wild: even from a lopsided population, the averages of samples pile up into a bell. Stats Without Walls lets you watch it: Learn > Sampling distribution simulator draws a thousand samples from any column, plots the thousand means, and compares their spread with the formula. The concept and the key number are fully visible here.

Stats Without Walls (click this path)

```
Learn > Sampling distribution simulator
  Column: quiz_score  Statistic: mean  Sample size 16  Samples 1000
Exploration > Descriptives (select "quiz_score")
  tick: Std. error
```

What you'll see

N	16	
Std. deviation	1.15	
Std. error of Mean	0.29	(this is SD / sqrt(N))

■ READ THE OUTPUT

The **standard error** is the standard deviation divided by the square root of the sample size, and the app prints it for you: about $1.15 / \sqrt{16} = 0.29$. The simulator shows the same number a second way: the standard deviation of the thousand sample means lands right next to it. That single number is the beating heart of the chapter. It measures how much a sample mean would bounce around from sample to sample, and those bouncing means pile into a bell. Larger samples make the square root bigger, so the standard error shrinks.

■ YOUR TURN

The standard error is $SD / \sqrt{N}$. Without touching the app, predict what happens to it if the sample size grows from 16 to 64. Bigger or smaller? By roughly what factor?

Chapter 8

How Sure Are We?

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Build a confidence interval for a mean and for a proportion, and read it the way the chapter taught.

A confidence interval is a range of believable values for the truth. Stats Without Walls hands you the whole interval.

Stats Without Walls (click this path)

```
T-Tests > One Sample T-Test (select "quiz_score")  
under "Additional Statistics" check: Confidence interval
```

```
Frequencies > 2 Outcomes (Binomial test) ("response")  
the proportion and its confidence interval appear
```

What you'll see

Mean 95% CI:	13.4 to 14.6
Proportion (Yes):	0.45
Proportion 95% CI:	0.30 to 0.61

■ READ THE OUTPUT

The mean interval, about 13.4 to 14.6, says: we are 95% confident the true average score sits in there. The proportion interval, about 0.30 to 0.61, does the same for the true “yes” rate (18 of 40 said yes, which is 0.45). Want 90% instead of 95%? Change the confidence level in the options and watch the interval shrink.

■ YOUR TURN

Change the confidence level for the mean from 95% to 99%. Wider or narrower? Why does being *more confident* cost you a *wider* net?

Chapter 9

Testing What We Believe

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Run a one-sample hypothesis test for a mean and for a proportion, and find the p-value.

Chapter 9 is the logic of the p-value. Stats Without Walls runs the test and prints the p-value plainly; your job is to decide what it means.

Stats Without Walls (click this path)

```
T-Tests > One Sample T-Test (select "quiz_score")
  set "Test value" = 13

Frequencies > 2 Outcomes (Binomial test) ("response")
  set "Test value" = 0.5
```

What you'll see

```
One Sample T-Test
t = 3.46    df = 15    p = 0.003
```

■ READ THE OUTPUT

Find the p in the output. Here it is 0.003, which is smaller than 0.05, so we reject the claim that the mean is 13; the evidence says the true mean is higher. Same move for the proportion test: set the Test value to 0.5, read the p-value, compare it to your significance level, decide. Stats Without Walls never decides for you, and that is the point of the chapter.

■ CHECK YOUR VOICE

When the results panel fills up, the anxious voice says “I don’t know what any of this means.” You do. You are looking for one number, the p-value, and asking one question: smaller than 0.05 or not? Everything else is supporting detail.

Chapter 10

Comparing Two Groups

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Compare two group means (independent and paired) and two proportions.

Chapter 10 asks: is the difference between two groups real, or just noise? One menu per situation.

Stats Without Walls (click this path)

T-Tests > Independent Samples T-Test

Dependent: score Grouping: group (two_groups.csv)

T-Tests > Paired Samples T-Test

Paired: before and after (before_after.csv)

Two proportions:

Frequencies > Contingency Tables (chi-square)

What you'll see

Independent Samples T-Test

Group means: A = 78.2 B = 66.5

t = 2.49 p = 0.036

■ READ THE OUTPUT

The two group means are about 78 and 67, and the p-value 0.036 is below 0.05, so the gap is bigger than noise would easily explain. Choosing **Independent Samples** versus **Paired Samples** is the whole difference between “two separate groups” and “the same people measured twice.” That decision is the chapter’s key move; in Stats Without Walls it is just picking the right menu. For two proportions, the Contingency Tables chi-square is the equivalent test.

■ YOUR TURN

Which design, independent or paired, matches a weight-loss study that weighs the *same* people in January and June? Which menu would you click for it?

Chapter 11

When Three or More Groups Disagree

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Run a one-way ANOVA and, if it is significant, find *which* groups differ.

When there are three or more groups, you do not run many t-tests. You run one ANOVA. Chapter 11's whole method is one menu in Stats Without Walls.

Stats Without Walls (click this path)

```
ANOVA > One-Way ANOVA
```

```
Dependent Variable: score      Grouping Variable: group
```

```
open "Post-Hoc Tests" and check: Tukey
```

What you'll see

```
One-Way ANOVA
```

```
F = 15.44      p = 0.004
```

```
Tukey post-hoc: compares A vs B, A vs C, B vs C
```

■ READ THE OUTPUT

Read the p-value: 0.004, below 0.05, so at least one group's mean really differs. But ANOVA does not say which one. Turning on **Tukey** post-hoc then compares each pair (A vs B, A vs C, B vs C) and flags the real gaps. That two-step logic, "is there a difference, then where," is the heart of the chapter.

■ YOUR TURN

Look at the Tukey table. Which pair of groups has the smallest p-value? That is the pair that differs the most.

Chapter 12

Counts, Categories, and Surprises

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Run both chi-square tests: goodness-of-fit (one variable) and independence (two variables).

Chapter 12 is about counts that surprise us. Are the categories as evenly split as expected? Are two categorical variables related? Two flavors of one test.

Stats Without Walls (click this path)

Goodness of fit (**is** the die fair?):

Frequencies > N Outcomes (chi-square goodness of fit)

Variable: face Counts: **count** (dice60.csv)

Independence (are caffeine and sleep related?):

Frequencies > Contingency Tables (the Chapter 5 **table**)

What you'll see

N Outcomes chi-square (goodness of fit)

X-squared = 2.8 df = 5 p = 0.73

■ READ THE OUTPUT

For the die, the p-value 0.73 is large (well above 0.05), so we do *not* have evidence the die is unfair; the ups and downs are ordinary luck. For the caffeine-and-sleep table, a small p-value would say the two variables are related, not independent. Same idea, two questions, exactly as the chapter frames them.

■ YOUR TURN

Change the die counts to something lopsided, like 2, 3, 4, 20, 6, 25, and re-run. Does the p-value drop below 0.05 now? What would you conclude about that die?

Chapter 13

Drawing the Line

■ WHAT THIS CHAPTER DOES IN STATS WITHOUT WALLS

Measure correlation, fit a regression line, read its slope and R^2 , and predict.

The last chapter draws the line through a scatterplot and asks what it means. Stats Without Walls gives you the correlation, the line, and the numbers to make a prediction.

Stats Without Walls (click this path)

```
Regression > Correlation Matrix
  select: hours and grade          (study_hours.csv)

Regression > Linear Regression
  Dependent Variable: grade      Covariate: hours
  open "Model Fit" for R-squared; "Model Coefficients" for the line
```

What you'll see

Correlation r	= 0.9985
Intercept	= 46.47
Slope (hours)	= 5.69
R-squared	= 0.997

■ READ THE OUTPUT

$r = 0.9985$ is a very strong positive relationship. The hours slope 5.69 means each extra study hour adds about 5.7 points. $R\text{-squared} = 0.997$ says hours explain about 99.7% of the variation in grade (unusually clean, because this is teaching data). To predict a grade at 3.5 hours, plug into the line: $46.47 + 5.69 \text{ times } 3.5$, about 66.4 points. Slope, strength, prediction: the whole chapter, on one panel.

■ CHECK YOUR VOICE

You just fit a regression model, the thing that sounded impossible on day one. Notice that. The clicks got simpler as the ideas got bigger, because the computer carries the arithmetic. You carried the understanding the whole way.

The Datasets, Chapter by Chapter

Every example in this book comes with a ready-made data file, so you can load the real numbers instead of typing them. All nine files live in a folder called `data`, sitting right next to this PDF. They are plain `.csv` files, which means they also open in **StatCrunch**, **R**, Excel, Google Sheets, and (typed into lists) the **TI-84**. One set of data, every tool.

How to load a dataset (do this once per file)

In Stats Without Walls, open the file from the Data menu, then read the type under each column name:

Stats Without Walls (click this path)

Data > Open CSV file > `data/class.csv`

Check the type under each column name

nominal or ordinal (words), continuous (numbers)

To change one: Data > Set variable type

The data fills the table on the left. If a variable is missing from a menu, it is almost always a type that needs fixing: a number column read as nominal because one cell holds text, or a category coded as numbers and read as continuous.

The nine datasets

File	Chapters	What's inside (columns)
class.csv	1, 3, 4, 8, 9	16 students: student, study_spot (Home/Library/Cafe), quiz_score
population40.csv	2	40 students to sample from: student_id, gpa
caffeine_sleep.csv	5, 12	100 people: caffeine (Yes/No), slept_well (Yes/No)
survey40.csv	8, 9	40 people: person, response (Yes/No; 18 said Yes)
two_groups.csv	10	Two independent groups: id, group (A/B), score
before_after.csv	10	Same people twice: student, before, after
three_groups.csv	11	Three groups for ANOVA: id, group (A/B/C), score
dice60.csv	12	A die rolled 60 times: face (1 to 6), count
study_hours.csv	13	hours studied and the grade earned (6 students)

Two chapters need no file. Chapter 6 (Predictable Randomness) uses the built-in **distrACTION** normal and binomial calculators, and Chapter 7 (From Sample to Truth) is taught by concept and by the standard error you read off Descriptives. Nothing to download for those two.

Loading each one is the same move. Data > Open CSV file, then check the type under each column name. For example, to rebuild the Chapter 5 two-way table, open `caffeine_sleep.csv`, check both columns read nominal, then go to Frequencies > Contingency Tables. To run the Chapter 10 two-group test, open `two_groups.csv`, set score to Continuous and group to Nominal, then go to T-Tests > Independent Samples T-Test.

Notice the pattern: get the variable types right, and the correct analysis is always waiting for you in the menu bar. That is the same file, read the Stats Without Walls way.

Quick Reference: The Whole Book in Stats Without Walls

One click-path per idea

Idea (chapter)	Stats Without Walls
Make categories (1)	Type under the column name; Data > Set variable type
Random sample (2)	Learn > Sampling distribution simulator; check with Descriptives
Bar plot / histogram (3)	Exploration > Descriptives > Plots
Center & spread (4)	Exploration > Descriptives > Statistics
Two-way table (5)	Frequencies > Contingency Tables
Normal / binomial (6)	distrACTION > Normal / Binomial
Sampling distribution (7)	Learn > Sampling distribution simulator; Descriptives: Std. error
Confidence interval (8)	T-Tests > One Sample T-Test (CI)
One-sample test (9)	One Sample T-Test; 2 Outcomes Binomial
Two groups (10)	T-Tests > Independent / Paired Samples
ANOVA (11)	ANOVA > One-Way ANOVA (Tukey)
Chi-square (12)	Frequencies > N Outcomes / Contingency Tables
Regression (13)	Regression > Correlation Matrix / Linear Regression

You did statistics, and you did it by pointing and clicking. That is not a small thing. Keep this page next to you; it is the whole course in thirteen click-paths.